

Multiple Choice (10 marks)

1. Millions of people living all over the world have cancer. Is cancer a pandemic?
 - (a) No, because cancer is not contagious.
 - (b) No, because cancer is not always fatal.
 - (c) Yes, because millions of people have cancer.
 - (d) Yes, because people all over the world have cancer.
2. City administrators can encourage energy conservation by
 - (a) lowering parking fees
 - (b) building larger parking lots
 - (c) decreasing the cost of gasoline
 - (d) lowering the cost of bus and subway fares
3. Which of the following is an example of a physical change?
 - (a) lighting a match
 - (b) breaking a glass
 - (c) burning of gasoline
 - (d) rusting of iron
4. A building is found to be losing heat to the outside during the winter months. Which would be the best solution to correct this design problem?
 - (a) increase amount of heating
 - (b) increase surface area of walls
 - (c) increase number of windows
 - (d) increase thickness of insulation
5. The heart is the major organ of the circulatory system. Which part of the body is responsible for delivering de-oxygenated blood to the heart?
 - (a) arteries
 - (b) capillaries
 - (c) valves
 - (d) veins

True / False (10 marks)

Answer True or False

6. True or False: Both an apple seed and a fertilized egg contain an embryo during their early life cycle stages.
7. True or False: Sexual reproduction produces offspring that are identical to their parents, maintaining uniformity in appearance.
8. True or False: Antibodies produce proteins that directly engulf and digest invading viruses.
9. True or False: Water is less dense as a solid than as a liquid, allowing ice to float and providing insulation for aquatic life in winter.
10. True or False: The increase in the number of homes built has likely contributed to the decline of the Miami Blue butterfly population in South Florida.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how electromagnetic energy can be utilized as the sole power source for a robot, including the principles of electromagnetism and potential applications in engineering.
12. Discuss the renewable energy resources available in Las Vegas for electricity generation, focusing on hydroelectric power and its environmental impacts. How can these resources be utilized sustainably?

End of Paper